# ITCS215
# Introduction to Interactive Multimedia Technology

Class: 1-2025
Module III: Introduction to Computer Graphics, Generative AI, and Audio
Lecture:  TUE 13:00-16:00

## Unit 1: Introduction to Multimedia in Module III

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# Teaching Modules

## ITCS215 Introduction to Interactive Multimedia Technology

- Module 1 (5 weeks)**:** Introduction to Image Analysis and Machine Learning.
- Module 2 (5 weeks): Introduction to Animation, Interactive Application, Text, and Font.
- **Module 3** (5 weeks): **Introduction to Computer Graphics, Generative AI, Sound-Visualization/Analytics**.

# Grading

## ITCS215 Introduction to Interactive Multimedia Technology

- Overall Score Parts
  - Class activities + Assignment 30%
  - Quizzes 30%
  - Final Exam 40%
- Module 1 (5 weeks)**:** Introduction to Image Analysis and Machine Learning.
- Module 2 (5 weeks): Introduction to Animation, Interactive Application, Text, and Font.
- **Module 3** (5 weeks): **Introduction to Computer Graphics, Generative AI, Sound-Visualization/Analytics**.
  - Class activities + Assignment 10%
  - Quizzes 10%
  - Final Exam 40/3 = 13.33%

# Module III: Outline

## ITCS215 Introduction to Interactive Multimedia Technology

- Introduction to Interactive Multimedia Technology Module II Syllabus
- Introduction to Types of Multimedia
- Quiz 1 (paper-based - NO AI)

# Syllabus

## Unit 1: Introduction to Multimedia in Module III

- Introduction to Types of Multimedia
- Overview of Computer Graphics
- Overview of Generative AI in Multimedia
- Overview of Sound and Visualization
- Overview of the Analytics on Computer Graphics, Generative AI, and Sound

# Introduction to Types of Multimedia

# Multimedia

## Unit 1: Introduction to Multimedia in Module III

- Multimedia refers to the integration of multiple forms of media, such as
    - text,
    - graphics,
    - audio,
    - video,
    - animations, and
    - interactive elements

  to create a rich, engaging, and dynamic user experience.
- It is widely used in entertainment, education, advertising, gaming, virtual reality, and more.

# Multimedia vs. Traditional Media

## Unit 1: Introduction to Multimedia in Module III

- Multimedia can be distinguished from traditional media, such as motion pictures or movies, by
    - the scale of the production and
    - the possibility of audience interactivity or involvement.

# Multimedia vs. Traditional Media

## Unit 1: Introduction to Multimedia in Module III

- The scale of the production in multimedia:
    - is smaller and less expensive. How?
- The possibility of audience interactivity or involvement:
    - led to the definition of “interactive multimedia.”

# Multimedia Application

## Unit 1: Introduction to Multimedia in Module III

- Multimedia application
    - links several content forms, and
    - provides tools for the user to
        - navigate,
        - interact, and
        - communicate

      with the computer.

# Types of Multimedia

## Unit 1: Introduction to Multimedia in Module III

- Multimedia can be classified into *different types* based on the type of content used.
- Some of the most common types of multimedia are:
    - Text-based Multimedia
    - Image-based Multimedia
    - Audio-based Multimedia
    - Video-based Multimedia
    - Interactive Multimedia

# Types of Multimedia

## Unit 1: Introduction to Multimedia in Module III

- Text-based Multimedia
  - This type of multimedia includes text, fonts, and typography.
  - It is used to create documents, presentations, and web pages.
- Image-based Multimedia
  - This type of multimedia includes still images and photographs.
  - It is used for creating graphics, logos, and illustrations.
- Audio-based Multimedia
  - This type of multimedia includes sound, music, and voiceovers.
  - It is used for creating podcasts, music videos, and sound effects.

# Types of Multimedia

## Unit 1: Introduction to Multimedia in Module III

- Video-based Multimedia
  - This type of multimedia includes moving images and animations.
  - It is used for creating movies, TV shows, and animated videos.
- Interactive Multimedia
  - This type of multimedia includes games, simulations, and virtual reality.
  - It is used for creating educational content, training programs, and entertainment.

# Multimedia Elements

## Unit 1: Introduction to Multimedia in Module III

- Various components make up a multimedia production.
- In a program, more than one media element is necessary to call it multimedia.
- For example:
    - Text and typography,
    - Coloured, pictorial, or graphical backgrounds,
    - Images and graphics such as photographs, illustrations, cartoons, diagrams, shapes, arrows, lines, etc.
    - 2D and 3D animation,
    - Audio files such as a song, sound effects, music, etc.
    - Video files such as a short movie clip, an advertisement, etc.
    - Buttons and a menu for easy navigation in a program.

# Multimedia Elements

## Unit 1: Introduction to Multimedia in Module III

- Multimedia is composed of various elements, each with its unique function.
- The components of multimedia are:
    - Text
    - Image
    - **Audio**
    - Video
    - **Computer Graphics-based Generated**
        - Computer-Generated Imagery (CGI)

# Computer-Generated Imagery (CGI)

## Unit 1: Introduction to Multimedia in Module III

- Computer-generated imagery, or CGI, is the application of the field of **computer graphics,** or more specifically, **special effects**.
- The CGI elements, for example, are
    - 2D Computer Graphics and Animation
    - 3D Computer Graphics and Animation
    - Visual Effect
    - Sound Effect

# Interactive Elements

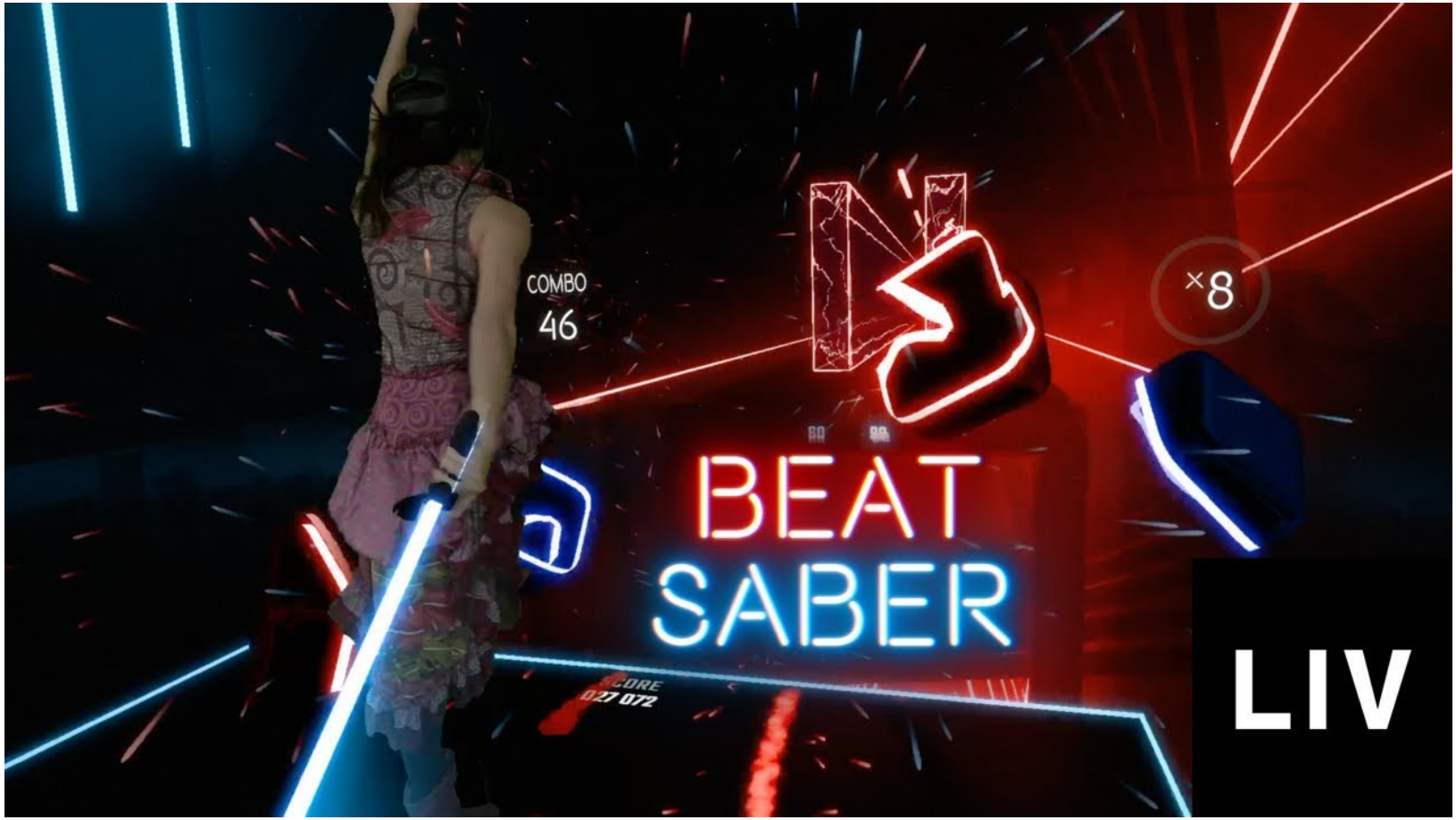
## Unit 1: Introduction to Multimedia in Module III

- The interactive elements include:
    - voice command,
    - mouse manipulation,
    - text entry,
    - touchpad,
    - touchscreen,
    - video capture of the user,
    - live participation - in live presentations.

# Quiz I

# Quiz I

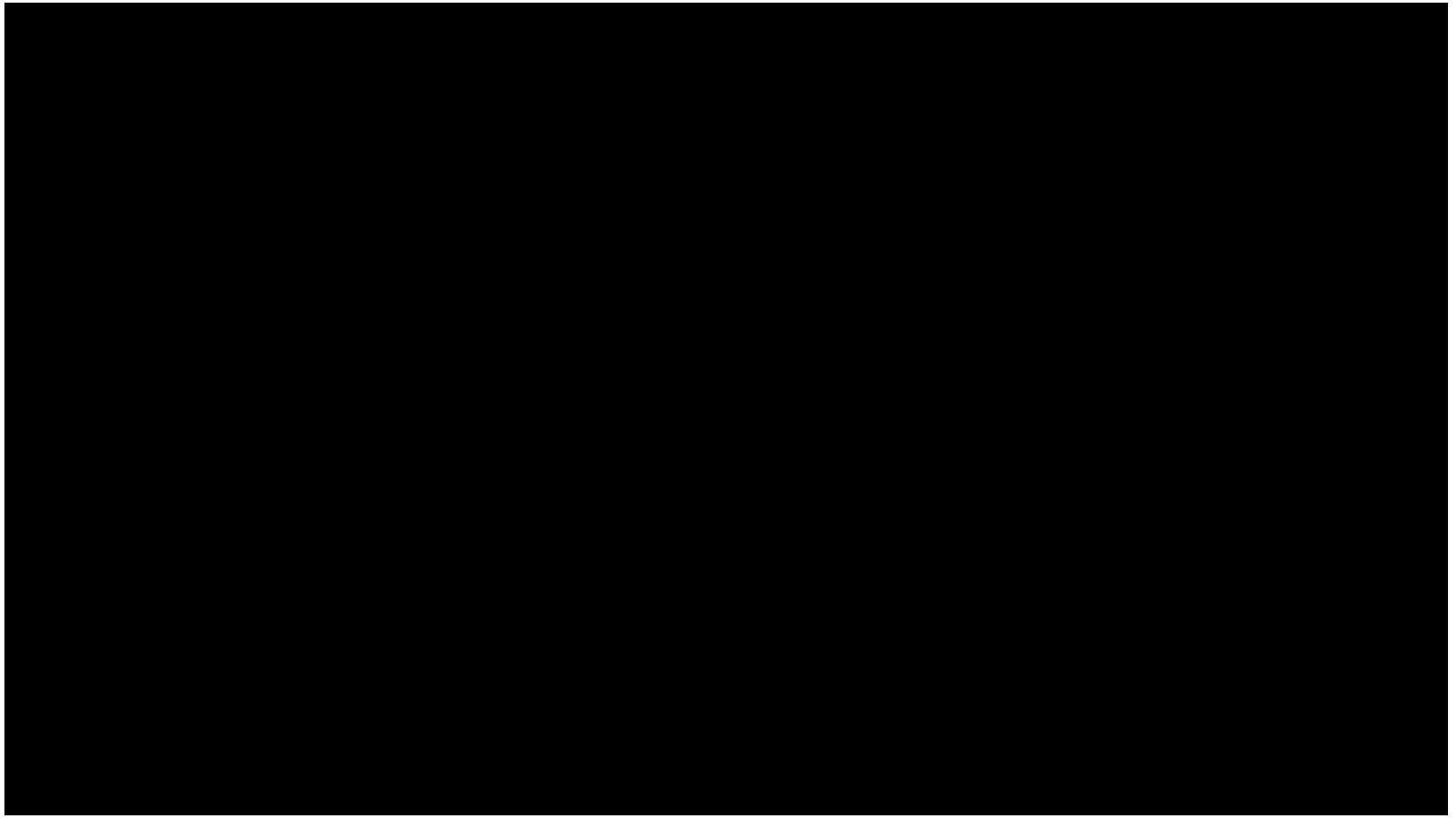
## Unit 1: Introduction to Multimedia in Module III




source: https://www.youtube.com/watch?v=gV1sw4lfwFw

# Quiz I

## Unit 1: Introduction to Multimedia in Module III



source: https://www.youtube.com/watch?v=gV1sw4lfwFw

# Quiz I

## Unit 1: Introduction to Multimedia in Module III

- Beat Saber is a kind of …….. multimedia.
- Beat Saber combines what media elements?
- What are the CGI elements found in this Beat Saber game?
- Please explain why the production of Beat Saber is less expensive compared to the traditional games (with a similar concept).
- What are the rules of the interaction?

